# Genuine Multipartite Entanglement Analysis — Molecular System

## MOLECULE

Methanamide (CHONH2)
Basis set : cc-pVDZ
Charge : 0
Multiplicity : 1
Total e- : 24
n-body order : 1

## GROUND STATE ENERGIES

E0 (Ha) : -168.96056339
E1 (Ha) : -168.72195263
Gap (Ha) : 0.23861076
HF (Ha) : -168.94623212

## CONVEX-ROOF METHODS

Methods run : gcg
Restarts : 8

C_LB : coherence witness bound
C_expo: unitary opt (exp map)
C_stf : Stiefel manifold (Alg. B)
C_UB : eigendecomposition bound

## ACTIVE SPACE CAS(6,5)

Frozen core orbs : 9
Frozen e- : 18
Active e- : 6
Active orbs : 5
Spin-orbitals : 10
Hilbert dim : 100

## SPIN PROPERTIES

<S2> : 0.000000
S : 0.0000
<Sz> : 0.000000
Multiplicity : 1
Spin state : Singlet (S=0)

## GME CRITERIA

C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
C_LB <= C_GME <= C_UB

## DOUBLE OCCUPANCY

<D> total : 2.988948
<D> / site : 0.597790

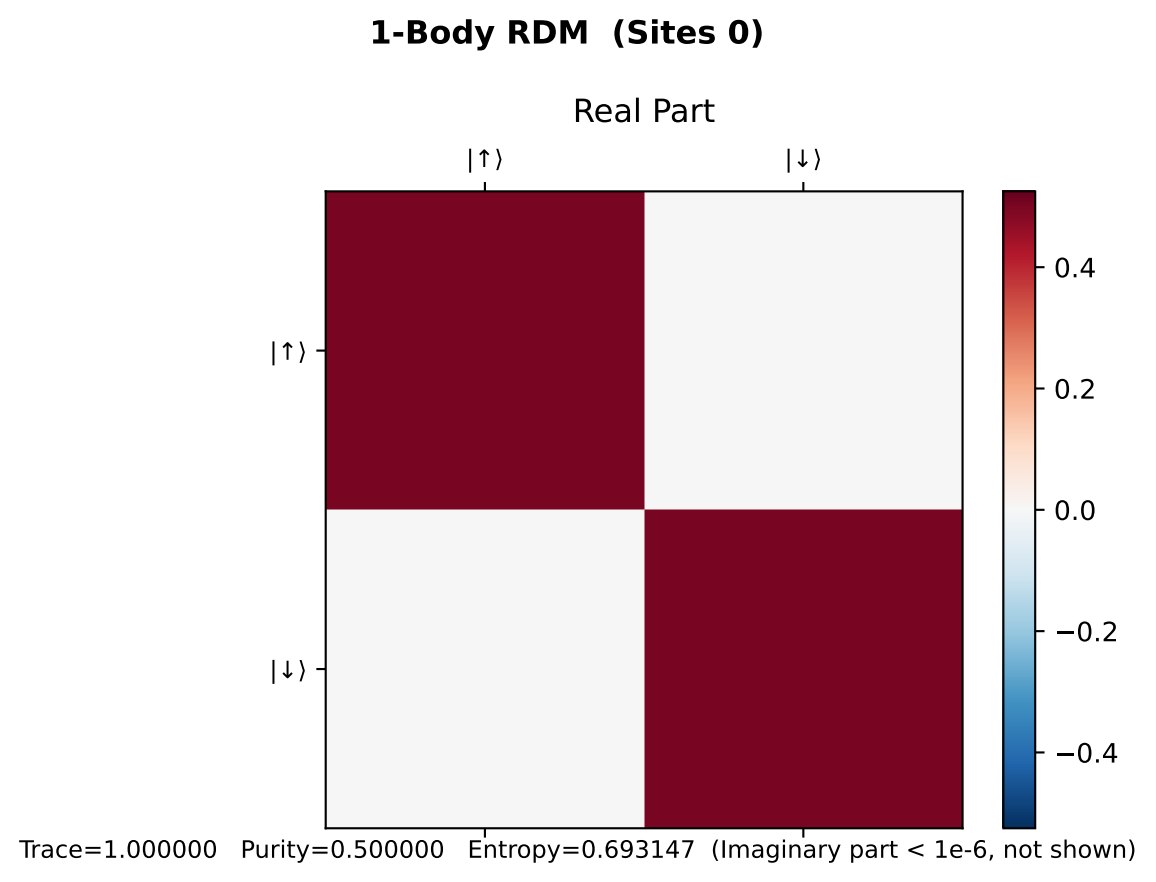
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Body order : 1
Clusters : 5

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## Summary — 1-Body RDMs  (Methanamide, CAS(6,5))

| Orbitals | Purity | C_GME (pure) | C_LB | C_gcg | C_stiefel | C_UB | State |
|---|---|---|---|---|---|---|---|
| 0 | 0.500000 | — | — | — | — | — | MIXED |
| 1 | 0.500000 | — | — | — | — | — | MIXED |
| 2 | 0.500000 | — | — | — | — | — | WEAKLY MIXED |
| 3 | 0.500000 | — | — | — | — | — | MIXED |
| 4 | 0.500000 | — | — | — | — | — | MIXED |


1-Body RDM (Sites 0)
Real Part
|↑⟩
|↓⟩
|↑⟩
|↓⟩
0.4
0.2
0.0
−0.2
−0.4
Trace=1.000000 Purity=0.500000 Entropy=0.693147 (Imaginary part < 1e-6, not shown)

# Analysis Summary - 1-Body RDM (Sites 0)

RDM Metrics
==========

| | |
|---|---|
| Trace(rho) | : 1.0000000000 |
| Purity Tr(rho^2) | : 0.5000000000 |
| Von Neumann entropy | : 0.6931471806 |
| Eigenvalue min | : 5.000000e-01 |
| Eigenvalue max | : 5.000000e-01 |
| Imaginary part | : < 1e-6 (not shown) |

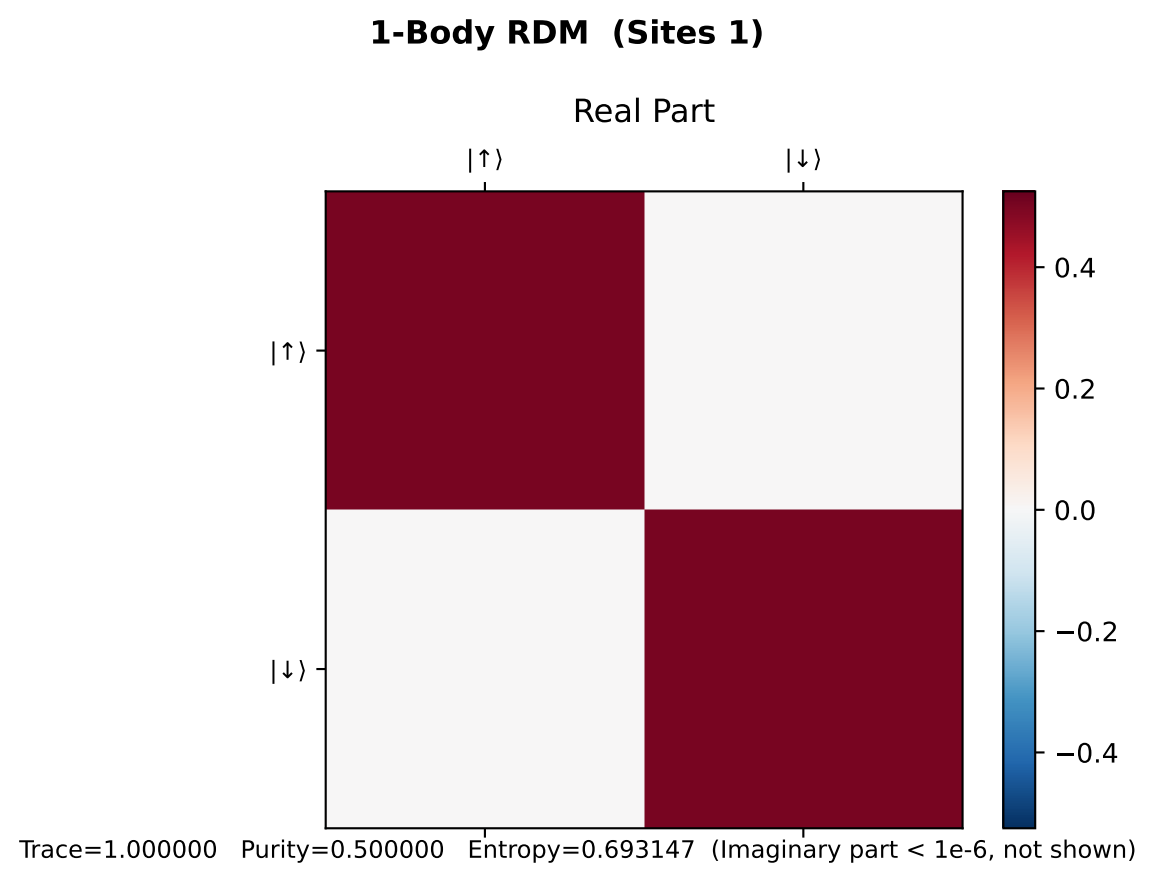
GME Analysis
==========

gamma = Tr(rho^2) = 0.50000000

| A | B | Tr(rho_A^2) |
|---|---|---|

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


1-Body RDM (Sites 1)
Real Part
|↑⟩
|↓⟩
|↑⟩
|↓⟩
0.4
0.2
0.0
−0.2
−0.4
Trace=1.000000 Purity=0.500000 Entropy=0.693147 (Imaginary part < 1e-6, not shown)

# Analysis Summary - 1-Body RDM (Sites 1)

RDM Metrics
==========

| | |
|---|---|
| Trace(rho) | : 1.0000000000 |
| Purity Tr(rho^2) | : 0.5000000000 |
| Von Neumann entropy | : 0.6931471806 |
| Eigenvalue min | : 5.000000e-01 |
| Eigenvalue max | : 5.000000e-01 |
| Imaginary part | : < 1e-6 (not shown) |

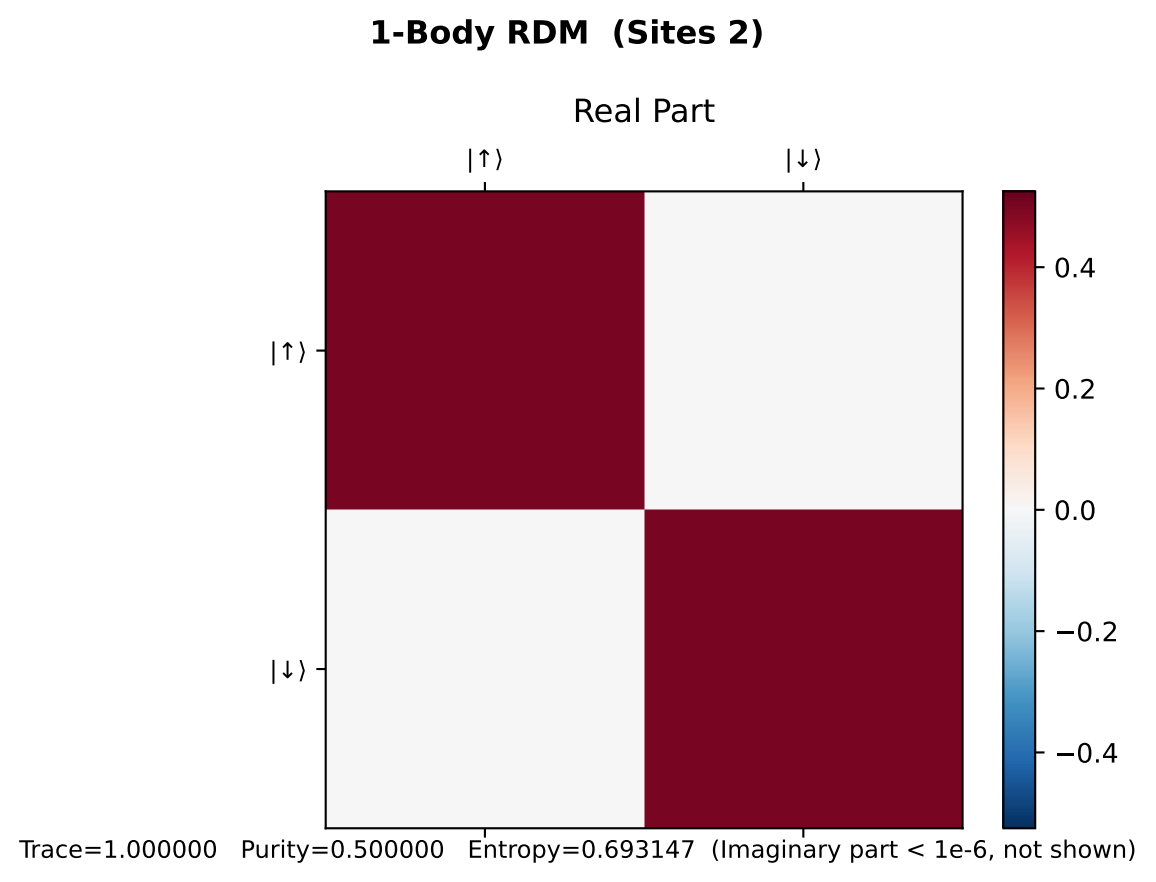
GME Analysis
==========

gamma = Tr(rho^2) = 0.50000000

| A | B | Tr(rho_A^2) |
|---|---|---|

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


1-Body RDM (Sites 2)
Real Part
|↑⟩
|↓⟩
|↑⟩
|↓⟩
0.4
0.2
0.0
−0.2
−0.4
Trace=1.000000 Purity=0.500000 Entropy=0.693147 (Imaginary part < 1e-6, not shown)

# Analysis Summary - 1-Body RDM (Sites 2)

RDM Metrics
==========

| | |
|---|---|
| Trace(rho) | : 1.0000000000 |
| Purity Tr(rho^2) | : 0.5000000000 |
| Von Neumann entropy | : 0.6931471806 |
| Eigenvalue min | : 5.000000e-01 |
| Eigenvalue max | : 5.000000e-01 |
| Imaginary part | : < 1e-6 (not shown) |

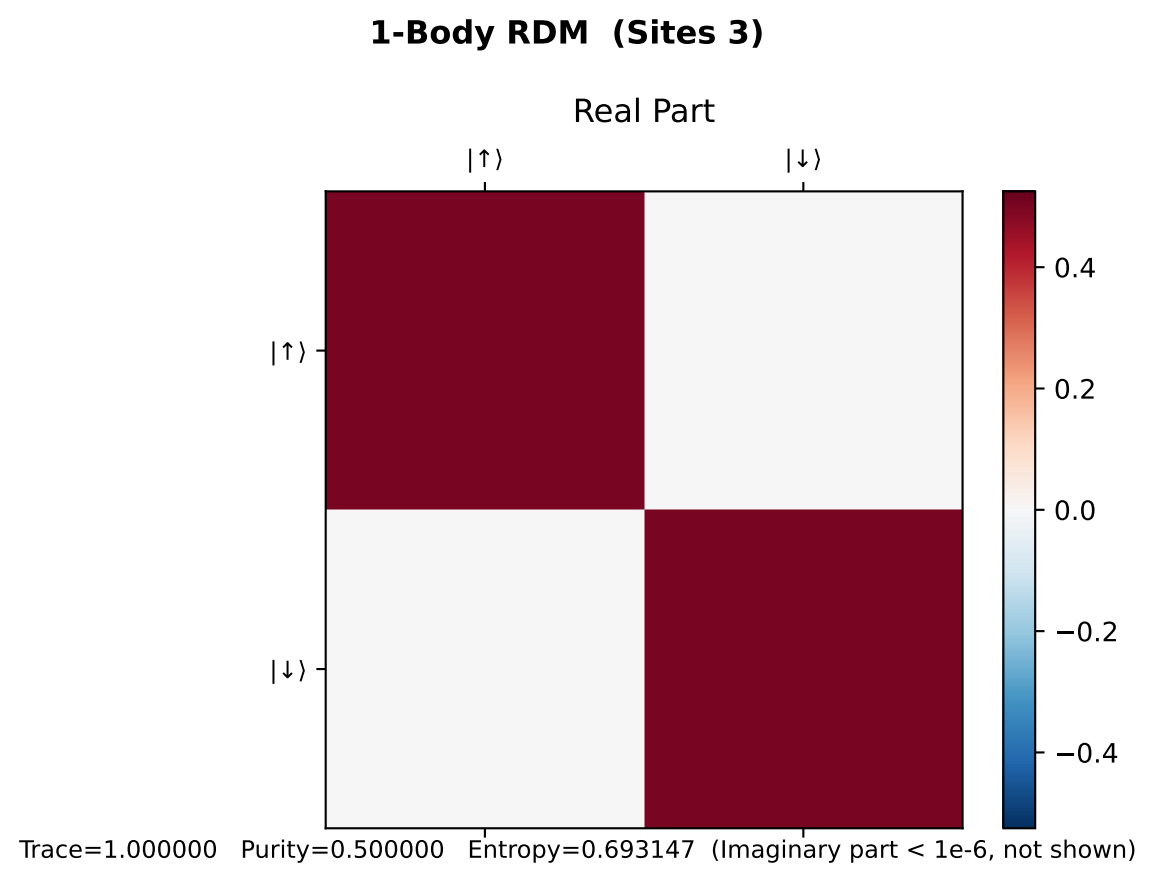
GME Analysis
==========

gamma = Tr(rho^2) = 0.50000000

| A | B | Tr(rho_A^2) |
|---|---|---|

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


1-Body RDM (Sites 3)
Real Part
|↑⟩
|↓⟩
|↑⟩
|↓⟩
0.4
0.2
0.0
−0.2
−0.4
Trace=1.000000 Purity=0.500000 Entropy=0.693147 (Imaginary part < 1e-6, not shown)

# Analysis Summary - 1-Body RDM (Sites 3)

RDM Metrics
==========

| | |
|---|---|
| Trace(rho) | : 1.0000000000 |
| Purity Tr(rho^2) | : 0.5000000000 |
| Von Neumann entropy | : 0.6931471806 |
| Eigenvalue min | : 5.000000e-01 |
| Eigenvalue max | : 5.000000e-01 |
| Imaginary part | : < 1e-6 (not shown) |

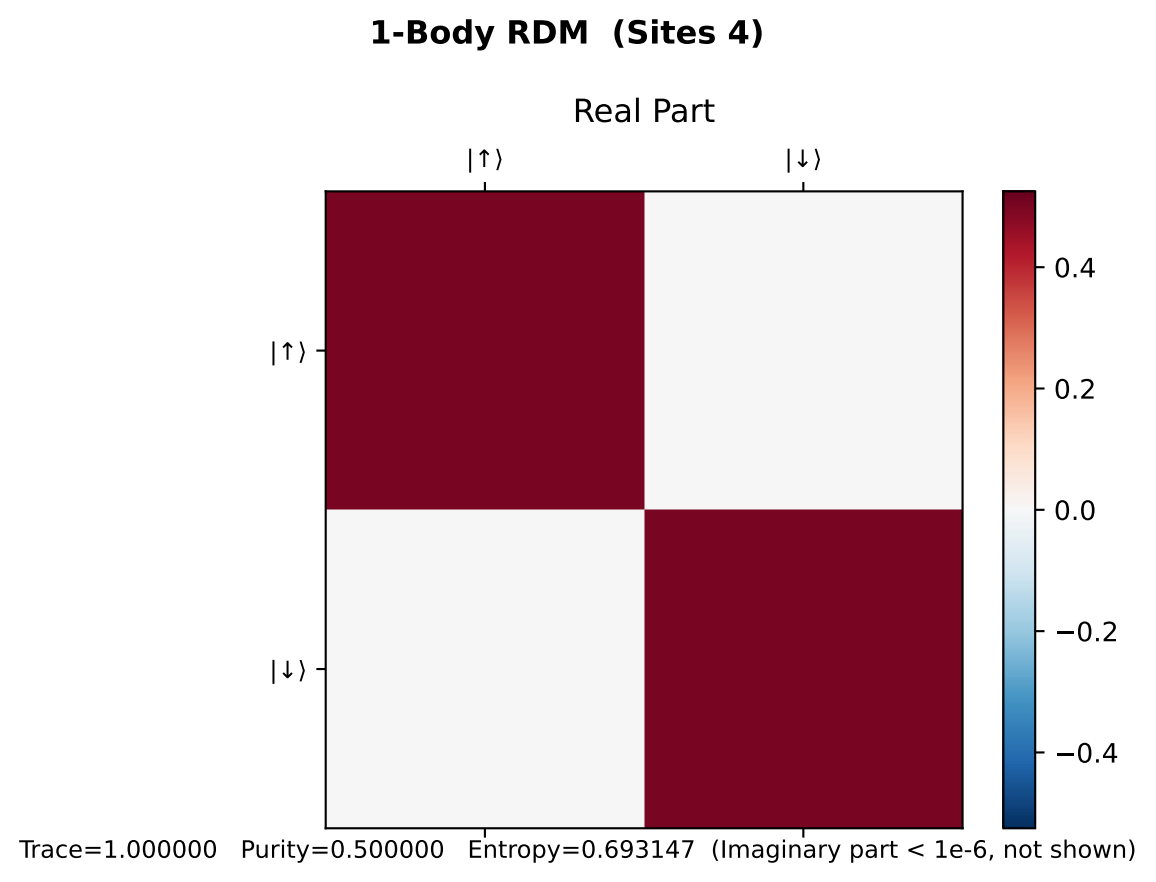
GME Analysis
==========

gamma = Tr(rho^2) = 0.50000000

| A | B | Tr(rho_A^2) |
|---|---|---|

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


1-Body RDM (Sites 4)
Real Part
|↑⟩
|↓⟩
|↑⟩
|↓⟩
0.4
0.2
0.0
−0.2
−0.4
Trace=1.000000 Purity=0.500000 Entropy=0.693147 (Imaginary part < 1e-6, not shown)

# Analysis Summary - 1-Body RDM (Sites 4)

RDM Metrics
==========

| | |
|---|---|
| Trace(rho) | : 1.0000000000 |
| Purity Tr(rho^2) | : 0.5000000000 |
| Von Neumann entropy | : 0.6931471806 |
| Eigenvalue min | : 5.000000e-01 |
| Eigenvalue max | : 5.000000e-01 |
| Imaginary part | : < 1e-6 (not shown) |

GME Analysis
==========

gamma = Tr(rho^2) = 0.50000000

| A | B | Tr(rho_A^2) |
|---|---|---|

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.